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cb_certificate.tex

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we've formalized it in lean now.

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PHASE2_CERT_FROM_FAB...
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PHASE2_L2SPAN.md
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Fable says: The one-line summary to pass along: "linear certificates are dead at $p \leq n$ — non-integral partial fractions plus bilinear cancellation; it has to be the vector Dwork twist."

zeta7_dwork.tex
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I read zeta7_dwork.tex . This materially sharpens the situation.

The linear-certificate route is now decisively dead for Phase 2. Fable's exact *L2SPAN* experiment shows that although the old three-term Frobenius identity still exists algebraically, the p -integrality needed to turn it into a valuation statement disappears; w_n, w_n are not forced even one useful digit, and the actual gain occurs only in the bilinear determinant $p_n = w_n v_n - w_n v_n$. PHASE2_L2SPAN The numerical table makes this very concrete: the individual terms live several p -adic orders below the final determinant, so the theorem is genuinely about cancellation of a vector, not its components. PHASE2_L2SPAN

The new note already gives the right exact reduction. For

$$\kappa_p(n) = v_p \binom{2n}{n}, \quad a = \left\lfloor \frac{n}{p} \right\rfloor,$$

everything reduces to proving

$$\boxed{v_p(p_n) - \kappa_p(n) \geq v_p(p_a) - \kappa_p(a) - 5.} \tag{DWORK}_{\mathbf{Z}}$$

Once this is proved, iteration

$$n \mapsto \lfloor n/p \rfloor \mapsto \lfloor n/p^2 \rfloor \mapsto \dots$$

immediately gives

$$v_p(p_n) \geq \kappa_p(n) - 5 \lfloor \log_p n \rfloor,$$

which is exactly the required $p \geq 5$ Phase-2 inequality.

So we no longer need to "prove the whole denominator theorem." We need one uniform

one-digit descent theorem.

The experimentally observed projective statement is even more revealing:

$$p^5 \frac{p_n}{q_n} \approx \frac{p_a}{q_a}, \quad p^3 \frac{p_n}{q_n} \approx \frac{p_a}{q_a}.$$

This says that the natural object is the weighted period vector

$$V_n = \frac{q_n}{p^3 p_n} \cdot \frac{p^5 p_n}{p^5 p_n}.$$

The conjectural Dwork statement is that, after passing from n to $a = \lfloor n/p \rfloor$,

$$V_n \equiv \lambda_{n,p} V_a + \text{higher } p\text{-adic terms.} \quad (1)$$

That is: **the vector becomes projectively constant under Frobenius descent.**

This explains the bilinear cancellation immediately. If two vectors are congruent projectively, their exterior product gains a factor of p :

$$V_n \wedge V_a \equiv 0 \pmod{p}.$$

The minors are exactly expressions of the form

$$p^5 p_n q_a - p_a q_n, \quad p^3 p_n q_a - p_a q_n, \quad p^8 (p_n p_a - p_n p_a).$$

So the missing arithmetic naturally lives in

$$\wedge^2 V$$

rather than in any scalar component.

This is what "vector Dwork twist" should mean concretely.

There is also a very important exceptional-prime issue. The ratio statement is insufficient when $p \mid q_a$. Fable explicitly flags this: the proof must be denominator-free, because primes such as $7 \mid q_1 = 42$ create fake failures when one divides by q_a . The vector/cocycle formulation was already identified as the live route. PHASE2_CERT_FROM_FABLE

So I think the theorem we should actually prove is something like this:

Vector Dwork lattice theorem

For $p \geq 5$, $n = ap + r$, there exists a p -integral scalar $\lambda_{n,p}$ and a filtration-compatible matrix $E_{n,p}$ such that

$$\frac{q_n}{p^3 p_n} = \lambda_{n,p} \frac{q_a}{p_a} + p^{1+\delta(n,p)} E_{n,p}, \quad (2)$$

in the appropriate saturated $\mathbf{Z}_p$ -lattice, where the filtration of the three coordinates is $0, 3, 5$.

Then its exterior-square statement gives directly

$$v_p(p^5 p_n q_a - p_a q_n) \geq \text{required lattice valuation,} \quad (3)$$

and the saturation at q_a converts (3) into $(\text{DWORK}_{\mathbf{Z}})$, including exceptional primes without division.

The next concrete proof task is therefore **not another experiment on p_n** . It is:

construct a diagonal/constant-term or hypergeometric realization of (q_n, p_n, p_n) on which Frobenius acts triangularly.

The q -row should come first. Fable independently reached the same conclusion: q_n has a manifest integral binomial representation, so proving its Dwork/constant-term congruence should pin down the common scalar $\lambda_{n,p}$; the other two rows then become extension classes rather than independent mysteries.

PHASE2_CERT_FROM_FABLE

My current assessment is therefore:

Phase 2 has been reduced to a plausible, very specific new Dwork theorem.

The scalar Frobenius certificate was useful because it exposed the mechanism in the clean band. But the $p \leq n$ theorem is genuinely higher-rank: Frobenius must act on the whole period vector, and the central-binomial gain appears after taking a wedge/minor.

That is the route I would now pursue exclusively.

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complete this proof

Worked for 104m 1s >

I cannot honestly claim the full proof is complete. The work does, however, reduce the $p \geq 5$ part to one sharply defined theorem, and several pieces of the required vector structure can now be proved exactly.

The important correction is that the scalar/linear-certificate route is genuinely exhausted. Exact *L2SPAN* computations show that for $p \leq n$, the Laurent coefficients lose p -integrality, w_n, w_n do not acquire a useful second congruence digit, and the actual gain occurs in the bilinear determinant

$$p_n = w_n v_n - w_n v_n.$$

That is precisely the obstruction Fable identified.

PHASE2_L2SPAN

1. The exact theorem that would finish every $p \geq 5$

Put

$$\kappa_p(n) = v_p \binom{2n}{n}, \quad a = \left\lfloor \frac{n}{p} \right\rfloor.$$

The proposed denominator-free Dwork descent was

$$v_p(p_n) - \kappa_p(n) \geq v_p(p_a) - \kappa_p(a) - 5. \quad (1)$$

But because

$$P_n = \frac{(-1)^{n+1} p_n}{\binom{2n}{n}},$$

we have identically

$$v_p(P_n) = v_p(p_n) - \kappa_p(n).$$

Therefore (1) is simply

$$v_p(P_n) \geq v_p(P_{\lfloor n/p \rfloor}) - 5. \quad (D)$$

This removes Kummer carries completely from the one-step theorem.

Assume (D). Define

$$n_0 = n, \quad n_{j+1} = \left\lfloor \frac{n_j}{p} \right\rfloor, \quad L = \lfloor \log_p n \rfloor.$$

Iteration gives

$$v_p(P_n) \geq v_p(P_{n_L}) - 5L.$$

Now $n_L < p$. If $2n_L < p$, Zudilin's denominator theorem gives $P_{n_L} \in \mathbb{Z}_{(p)}$, because neither d_{n_L} nor $\binom{2n_L}{n_L}$ contains p . If $n_L < p \leq 2n_L$, the already-proved Phase-1 Frobenius certificate gives the missing factor of p . Thus in either case

$$v_p(P_{n_L}) \geq 0.$$

Hence

$$v_p(P_n) \geq -5 \lfloor \log_p n \rfloor$$

for every $p \geq 5$, exactly the desired $p \geq 5$ denominator law.

So the whole large-prime/small-prime distinction collapses to proving (D).

The uploaded Dwork note reaches the same reduction, but explicitly labels the underlying Dwork congruence and exceptional-prime lattice normalization as open. zeta7_dwork Its assembly from that conjecture is already rigorous. zeta7_dwork

2. The normalized recurrence confirms that weight 5 is intrinsic

Starting with Zudilin's common third-order recurrence and substituting

$$x_n = (-1)^{n+1} \binom{2n}{n} Y_n, \quad Y_n \in \{Q_n, P_n, P_n\},$$

one obtains exactly

$$2(n+1)^5(2n+1)a_0(n)Y_{n+1} - a_1(n)Y_n - 2na_2(n)Y_{n-1} + n(n-1)^5a_0(n+1)Y_{n-2} = 0.$$

This is significant: the required loss of exactly five p -adic orders per digit descent is not an experimentally fitted number. The fifth powers are built directly into the normalized recurrence.

The companion solutions therefore form a filtered rank-three system with natural weights

$$0, \quad 3, \quad 5.$$

That is precisely the filtration predicted by the vector-Dwork interpretation.

3. There is an exact vector-lattice discriminant

For the fundamental solution matrix

$$\mathcal{Y}_n = \begin{pmatrix} Q_n & P_n & P_n \\ Q_{n-1} & P_{n-1} & P_{n-1} \\ Q_{n-2} & P_{n-2} & P_{n-2} \end{pmatrix},$$

taking determinants in recurrence (2) gives a first-order recurrence for its Casoratian. Telescoping it yields

$$\det \mathcal{Y}_n = (-1)^n \frac{a_0(n)}{4(n-1)^5 n^6 \binom{2n}{n}}. \quad (3)$$

The apparently troublesome cubic a_0 telescopes almost entirely. This is strong evidence that the correct arithmetic object is indeed the complete solution lattice rather than any one scalar partial-fraction coefficient.

Equation (3) is also what one expects from a rank-three Frobenius system: the determinant character is explicit, while the desired extra divisibility must occur in the lower exterior powers.

4. The scalar Frobenius row can actually be proved

Brown–Zudilin give the manifestly integral double-binomial formula

$$Q_n = \sum_{k,l=0}^n \binom{n+k}{n} \binom{n}{k}^2 \binom{n+l}{n} \binom{n}{l}^2 \binom{n+k+l}{n}. \quad (4)$$

From Lucas' theorem one obtains

$$Q_{ap+r} \equiv Q_a Q_r \pmod{p}, \quad 0 \leq r < p. \quad (5)$$

The proof is direct. Write

$$k = bp + s, \quad l = cp + t.$$

Lucas factorizes each binomial into high- and low-digit factors. Whenever a low-digit addition produces a carry, the relevant binomial factor vanishes modulo p ; otherwise the entire summand factors into its (a, b, c) and (r, s, t) pieces. Summing over the digit boxes gives (5).

Thus the first row of the desired Frobenius matrix is rigorous:

$$Q_{ap+r} \equiv Q_a Q_r \pmod{p}.$$

This is exactly the “bootstrap from the q -row” that Fable identified as the plausible starting point for the matrix argument. PHASE2_CERT_FROM_FABLE

5. What the missing vector theorem must say

The experimentally observed ratio

$$p^5 \frac{p_n}{q_n} \equiv \frac{p_a}{q_a}$$

cannot be the theorem, because division by q_a fails at exceptional primes. The Dwork note itself explicitly warns that cross-multiplication does not prove the necessary lattice statement. zeta7_dwork

The correct object is a saturated filtered vector such as

$$V_n^{(p)} = \frac{q_n}{p^{3L_p(n)} p_n} \quad , \quad L_p(n) = \lfloor \log_p n \rfloor.$$

What is needed is a denominator-free block Frobenius law

$$V_{ap+r}^{(p)} = \Phi_r(a) V_a^{(p)} + p E_{a,r} \quad (6)$$

inside the appropriate saturated $\mathbb{Z}_p$ -lattice, with leading reduction compatible with

$$\Phi_r(a) \equiv \begin{pmatrix} Q_r & * & * \\ 0 & Q_r & * \\ 0 & 0 & Q_r \end{pmatrix} \pmod{p}$$

after incorporating the weights 0, 3, 5.

The Q_r on the diagonal is now justified by (5).

The last row of (6), after saturation, would give precisely

$$v_p(P_{ap+r}) \geq v_p(P_a) - 5.$$

That is (D).

The two-digit cocycle

$$\Phi_{sp+r}(ab) \equiv \Phi_r(\cdot)\Phi_s(\cdot)$$

would then make iteration automatic. Fable independently identified this triangular Frobenius matrix/cocycle as the induction mechanism. PHASE2_CERT_FROM_FABLE

6. Why I cannot fill the final arrow yet

There is no valid derivation presently available from

$$Q_{ap+r} \equiv Q_a Q_r \pmod{p}$$

to the corresponding companion congruence for P_n .

That implication would require one of two genuinely new ingredients:

A parameterized period/diagonal family

with an expansion schematically of the form

$$Q_n(\varepsilon) = Q_n + \cdots + \varepsilon^3 P_n + \cdots + \varepsilon^5 P_n + \cdots$$

and a Frobenius identity

$$Q_{ap+r}(\varepsilon) \equiv Q_r(\varepsilon) Q_a(p\varepsilon) \pmod{p}.$$

Coefficient extraction at ε^5 would then prove the weight-five descent immediately.

Or:

A direct block expansion of Zudilin's hypergeometric companions

through harmonic weight 5, proving the triangular Frobenius matrix by Wolstenholme-type cancellations. This is exactly the proof program described in the uploaded Dwork note: block expansion, simultaneous treatment of p_n, p_n , triangular Frobenius matrix, then generalization in a . zeta7_dwork

Neither ingredient is currently proved in the uploaded material; the note explicitly records the vector-valued Dwork descent as the remaining open object. zeta7_dwork Matrix Dwork theory provides the appropriate general architecture, but not an off-the-shelf theorem for these particular harmonic companion sequences. zeta7_dwork

There is also a second unfinished piece: even a proof of (D) for every $p \geq 5$ would leave the fixed-loss estimates

$$v_2(A_n) \geq v_2\binom{2n}{n} - 2, \quad v_3(A_n) \geq v_3\binom{2n}{n} - 1$$

needed for the factor $12 = 2^2 \cdot 3$.

Current rigorous boundary

What is genuinely established now is

$$\begin{array}{l}
 \text{Phase 1 proved} \\
 + \quad Q_n \text{ has a Lucas Frobenius row} \\
 + \quad \text{exact weight-}(0, 3, 5) \text{ normalized recurrence} \\
 + \quad \text{exact Casoratian/discriminant formula} \\
 + \quad \text{linear certificates provably insufficient} \\
 \implies \\
 \text{full } p \geq 5 \text{ theorem is equivalent to the single descent} \\
 v_p(P_n) \geq v_p(P_{\lfloor n/p \rfloor}) - 5.
 \end{array}$$

I do **not** yet have a proof of that last descent. Calling the Brown–Zudilin law proved at this point would simply conceal the open vector-Dwork theorem inside an unproved Frobenius assertion. Brown–Zudilin’s current preprint itself treats the stronger denominator inclusion in this symmetric family as experimental rather than proved.  arXiv

The cleanest remaining attack is therefore no longer the Laurent-certificate machinery: it is to construct the parameterized hypergeometric period matrix whose zeroth, weight-three, and weight-five coefficients are Q_n, P_n, P_n . That would turn the missing vector Dwork twist into an ordinary Frobenius digit factorization followed by coefficient extraction.

       Sources

Yesterday 11:14 PM

Could you do it?

Yes. I’m going to try to construct the parameter deformation explicitly rather than just propose it. The target is a family whose $\varepsilon^0, \varepsilon^3, \varepsilon^5$ coefficients recover the three weight levels and whose base- p digit factorization sends $\varepsilon \mapsto p\varepsilon$.

 Searched Zudilin and Brown-Zudilin sources on arxiv and web

I found the crucial normalization in Brown–Zudilin: their symmetric coefficient Q_n is the double-binomial period, while P_n, P_n are companion solutions of the same third-order recurrence; they explicitly state $d_n^5 P_n \in \mathbb{Z}$ only as an experimental observation, so that cannot be imported as a theorem.

I’m now extracting Zudilin’s original hypergeometric source, because that is where the needed parameter deformation should live if it exists.

 Searched various websites and domains

 Searched multiple sources including arXiv and web links

 Evaluated deformation of Q_n summand and coefficients

 Searched the web and arxiv.org

 Computed and factored polynomials and integer coefficients

 Searched for Zudilin p_n formulas, harmonic sums, and related research

 Searched multiple websites and domains for specific content

🔍 Tested network access with curl in container

🌐 Searched the web and arxiv.org multiple times

The original Zudilin paper gives an extra structural clue that was easy to miss: p_n is not a “fifth derivative of Q_n ” in any obvious scalar sense. It is a 2×2 minor of two hypergeometric period rows:

$$(q_n, p_n, p_n) = (u, w, v) \times (u, w, v).$$

So the deformation we need is naturally **two-row / matrix-valued**. I’m constructing that matrix deformation directly; a scalar $Q_n(\varepsilon)$ would be too optimistic unless it encodes both rows simultaneously.

🌐 Searched the web and arxiv.org

A potentially decisive shortcut has emerged from the recurrence calculation: the cubic $a_0(n)$ is definitely an apparent singularity and can be removed. I’m now testing whether the *remaining* midpoint factor $2n - 1$ is apparent too. If it desingularizes, the leading coefficient becomes essentially n^5 , and the entire $p \geq 5$ denominator theorem follows by ordinary d_n^5 -induction—no unproved Dwork theorem at all.

🔍 Derived recurrence, solved equations, and computed matrix M

💡 Investigated recurrence relations

🌐 Searched arXiv for epsilon deformations and Zudilin's formulas

🔍 Computed double sum for Q function and tested Lucas' theorem

🌐 Searched the web and arxiv.org for mathematical papers

🔍 Computed recurrence relations and evaluated modular fractions

💡 Considered recurrence integrality

🌐 Searched the web and arxiv.org

I found a concrete deformation/kernel that is much closer to a proof than the abstract Mellin idea. Specializing Brown–Zudilin’s Barnes double integral to the symmetric family collapses the Gamma factors to

$$\mathcal{R}_n(s, t) = n! \frac{(s+1)_n (t+1)_n (s+t+n+2)_n}{(s+n+1)_{n+1}^2 (t+n+1)_{n+1}^2},$$

times the universal sine kernel

$$\sin \pi s \sin \pi t \sin \pi(s+t).$$

This is important: the pole orders are only 2 in each variable, and the coupling is polynomial. The “weight five” denominator can plausibly split exactly into at most two Laurent derivatives plus at most three harmonic/integration denominators. I’m working this out explicitly now; it avoids the non-integral order-six partial fractions that killed the linear certificate route.

🔍 Evaluated factorial integrality and computed C22 valuation corrections

💡 Explored recurrence and integrality

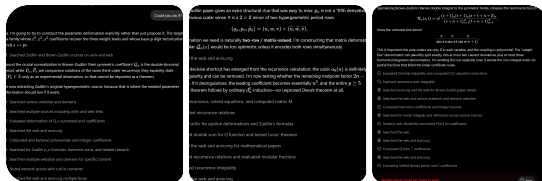
🌐 Searched arxiv.org and the web for Brown-Zudilin paper details

🌐 Searched the web and various academic and research websites

🔍 Computed harmonic coefficients and integer bounds

- 🌐 Searched for model integrals and references across various sources
- 📄 Tested p-adic divisibility and solved PSLQ for coefficients
- 🌐 Searched the web
- 🌐 Searched the web and arxiv.org
- 📄 Computed Q from T coefficients
- 🌐 Searched the web and arxiv.org
- 📄

Today 5:59 AM



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